13. Menus and Registers

Device settings and readings are organized as registers. The following tables show the registers by name and number, organized by menu. Registers that can be written are indicated by this symbol: ✎. Others are read-only and cannot be changed through the MENU or serial communications. Some registers can be accessed only through serial communications.

	Register Name	#	Register Contents		
COMMANDS MENU	COMMANDS ✎	8	Execute a command to control the device		
			-	0	(no commands, inactive)
			Reset	200H	Reset alarms
			Acknowledge	100H	Silence the horn, clear “service needed” relay
			Calibrate analyzer with gas, adjust settings, activate faults if response unacceptable		
			Zero Cal	300H	Perform zero calibration and adjust zero
			Span Cal	400H	Perform span calibration and adjust span
			Full Cal	500H	Perform zero and span calibration and adjust
			Test analyzer response with gases, but do not adjust the settings, ignore faults		
			Zero Test	600H	Test zero with gas, do not adjust
			Span Test	700H	Test span with gas, do not adjust
			Full Test	800H	Test zero and span, do not adjust
			Commands not on menu – serial port only		
			Save Data	900H	Write settings to nonvolatile EERPOM to save.
			Cold Start	0A00H	Load default values and write to EEPROM.
			Commands for security – not on menu		
			Lock	0F00H	Locks out calibrations, software lock.
			Unlock	1000H	Unlocks calibrations.

	Register Name	#	Register Contents
GAUGES MENU	READING	1	The hydrocarbon reading in parts per million carbon by volume (PPM) or milligrams carbon per normal cubic meter (MG)
	STATUS	2	Device status messages, codes, and their meanings **Message** / **Code** / **Meaning** Normal / 0 / Normal operation, no alarms Warning / 1 / Warning alarm, reading above warning level Danger / 2 / Danger alarm, reading above danger level Downscale / 3 / Fault, reading lower than –10%LFL Zero Cal / 4 / Zero calibration in progress Span Cal / 5 / Span calibration in progress Purging / 6 / Purging calibration gases Cal ends / 7 / Calibration ending Zero fail / 8 / Zero calibration failed, result unacceptable Span fail, low / 9 / Span calibration failed, result too low Span fail, high / 10 / Span calibration failed, result too high Warmup / 11 / Power-on time delay for self diagnostics RAM test / 12 / Self diagnostic of RAM RAM fail / 13 / RAM has failed self diagnostic test Data lost / 14 / Checksum for nonvolatile data in EEPROM failed EE fail / 15 / Self diagnostic check of EEPROM failed Vac Low / 16 / Flame cell vacuum too low Vac High / 17 / Flame cell vacuum too high Temp low / 18 / Temperature of analyzer is too low Temp high / 19 / Temperature of analyzer is too high A/D fail / 24 / Self check of A/D converter failed. TC2 Fail / 25 / Thermocouple for temperature control failed. Flameout / 26 / Sensing flame is out. ROM fail / 27 / Continuous self diagnostic of ROM has failed.
	PEAK HI	38	Highest reading taken since last reset command
	PEAK LOW	39	Lowest reading taken since last reset command
	HSG TEMP	9	Actual flame cell housing temperature in Degrees Centigrade
	SET TEMP ✎	24	Flame cell housing temperature set point in Degrees Centigrade Min: 60°C / Max: 210°C / Default: 200°C
	CELL VAC	10	Flame cell vacuum measured in inches of water or kilopascals
	SET VAC ✎	23	Flame cell vacuum set point measured in inches of water or kilopascals Min: -45” / -11.2kPa Max: -30” / -7.4kPa Default: -40” / -10kPa
	ASP PRES	33	Calculated pressure needed to maintain flame cell vacuum set point expressed in psi or kPa.
	FLAME	11	Flame temperature (Thermocouple TC1 reading) in Degrees Centigrade
	RAW RDNG	12	Continuously updated hydrocarbon reading expressed as VDC

	REGISTER NAME	#	REGISTER CONTENTS
CAL MENU	RAW ZERO	29	Hydrocarbon reading from zero calibration in VDC
	RAW SPAN	30	Hydrocarbon reading from span cal minus the raw zero, in VDC
	CAL TIME ✎	28	Duration of each calibration interval in minutes from 1 to 4
	CAL RDNG ✎	7	Setting to adjust to, which final reading gets adjusted by during span calibration
	SPAN VDC ✎	56	Setting for expected RAW SPAN voltage rise during span calibration
	SPEED ✎	20	Speed of response of the sensor **Speed Code** Low 1 Slowest setting Med 2 High 3 Fastest setting
	CAL LOCK	60	Indication of calibration lock. Read only. Hardware Lock Bit 0 Software Lock Bit 1
ALARMS	WARNING ✎	5	The warning alarm level, in the same format as reading
	DANGER ✎	6	The danger alarm level, in the same format as reading
	FULLSCLE ✎	3	Maximum reading in PPM or MG represented by 20mA Min. 10 Max. 32000 Default 100
	UNITS		Controls units of measure for the reading, cell vac, set vac and aspirator pressure **US CUST / METRIC** 00 / 01 Reading: PPM / MG CELL VAC: IN / KPA ASP PRES: PSI / KPA
	LATCHES ✎	17	Alarms that will latch until manually reset, versus those that will autoreset Format is identical to RELAYS register.
	FAILSAFE ✎	18	Relays that are failsafe (normally energized) and which de-energize upon activation of the alarm. Format is identical to RELAYS register.
	VERSION	257 to 288	Device identification, in ASCII text string. 'TYPE FID' Analyzer Model ' VER 4.80' Software version ' SNR650 ' Analyzer assembly number ' 99 999 ' Analyzer serial number
OUTPUTS MENU	RLY TEST ✎	45	Relays activated for test. Format is identical to RELAYS register.
	MA TEST ✎	44	Circuit test of milliamp output, in milliamps, increases output only, deactivates upon leaving menu.
	4 MA ADJ ✎	25	Circuit adjustment of milliamp output at the 0%FS (4 milliamp) point – factory set
	20 MA ADJ ✎	31	Circuit adjustment of milliamp output at the full scale (20 milliamp) point – factory set
	MA CAL ✎	26	Value for milliamp output during calibrations
	MA FAULT ✎	32	Value for milliamp output during faults
	CONTROL1 ✎	36	Command assigned to control input #1. Can be any from COMMANDS menu except 9,10.
	CONTROL2 ✎	37	Commands assigned to control input #2. Can be any from COMMANDS menu except 9,10.
	RLY CNFG ✎	58	Configuration setting for the HORN relay 00 Default setting. The relay is used for the HORN function. 01 AIR/FUEL function. The relay energizes (contact closed) when flamecell is at proper operating temperature. 02 OVERRIDE function. Relay is always energized (contact always closed).

	REGISTER NAME	#	REGISTER CONTENTS
COM PORT MENU	SET BAUD ✎	22	Default Baud Rate Setting
	BAUD RATE ✎	47	Device Baud Rate Currently In Use. Can differ from the default setting if autobaud is “on” 0010H 19200 0011H 9600 0012H 4800 0013H 2400 0014H 1200 0015H 600 0016H 300
	AUTOBAUD ✎	48	Control for automatic baud rate detection feature (vs. fixed baud rate setting) 0 Off Baud rate fixed, equals “SET BAUD” value 1 On Device automatically seeks proper baud rate
	ADDRESS ✎	21	Port address of device from 1 to 255
	ID NBR ✎	19	User-assigned identification number from 0 to 65535

REGISTER NAME	#	Register Contents **Note:** These registers are not accessible through any menu
RELAYS	4	Logical state of the control relays. This data matches the corresponding LED indicators. A bit of 1 = “active”, 0 = “inactive” Data is in binary format. Packed flags: 0000 0000 TISZ HFDW T Heater I Ignition S Span solenoid valve Z Zero solenoid valve H Horn relay F Fault relay D Danger relay W Warning relay
DUMMY	46	A register that performs no function, but can be used to debug serial communications
A/D REF	16	The reference voltage of the A/D converter. Should be 506 ± 25.

13.1. Register Conversion Factors

After it is acquired, the serial port data is converted into engineering units using a conversion factor. For the following registers, the data is a signed 16 bit word. Negative numbers are in two‘s complement form. To convert the register data, multiply by the conversion factor in the table.

REGISTERS	Format	Conversion Factor
READING PEAK HI PEAK LO CAL RDNG WARNING DANGER MA TEST MA CAL MA FAULT	Reading	Multiply by 0.0000390625 multiplied by FULLSCLE If FULLSCLE is less than 101 Then Reading has one decimal point If FULLSCLE is greater than 100 Then Reading has no decimal point
MA TEST MA CAL MA FAULT	Reading format expressed as milliamps	Multiply by 0.000625, then add + 4.0 Milliamps
HSG TEMP SET TEMP FLAME	Temperature	Multiply by 0.24426 °C
RAW RDNG RAW ZERO RAW SPAN SPAN VDC	Voltage	Multiply by 0.0024426 VDC
CELL VAC SET VAC	Flame cell vacuum	Multiply by 0.0351734 inches of water column Or Multiply by 0.00879336 kiloPascals
ASP PRES	Aspirator air pressure	Multiply by 0.105882 psi Or Multiply by 0.729412 kPa
4 MA ADJ 20 MA ADJ	Milliamps	Multiply by 0.1 Milliamps